

Paper ED-CCC — ED Identification of Conformal Cyclic Cosmology

A Generative Paper of the Event Density Program

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of CCC’s twistor mass-conservation theorem (Meissner–Penrose 2025 §III) from substrate primitives. The conservation integral is INHERITED; ED supplies a substrate-side identification, not a substrate-graph derivation.
2. It does **not** claim a substrate first-principles derivation of the Hawking-spot $\delta T/T \sim 10^{-3}$ value or of the conformal-time bridge η^G . Both are INHERITED from Meissner–Penrose 2025 + observational anchors.
3. It does **not** claim closed-proof reconstruction of CCC. CCC is in the conformal-GR derivation tradition; ED is in the substrate-ontology lineage (’t Hooft cellular-automaton; Wolfram Ruliad; causal-set program). The paper supplies structural correspondence, not reduction in either direction.
4. It does **not** introduce new substrate primitives. P-CCC-Identification is defined in §2.
5. It does **not** commit ED to cyclic cosmology. ED’s SCBU boundary is consistent with both cyclic-continuation (ED-CCC reading) and single-aeon-termination (ED-non-cyclic reading); the substrate-graph derivation of the post-boundary ignition mechanism is OPEN (audit row 17).
6. **Route A flag:** This paper consumes $\xi_{canonical}$ scale-correspondence content (Paper_092). Substrate derivation of $\ell_{V5}(H_0)$ and hence of $\xi_{canonical}$ from substrate parameters is **OPEN**. Quantitative consolidations downstream of SC-4.x inherit this OPEN status.

Abstract

CCC’s structural elements (Meissner–Penrose 2025) are identified with substrate-side content already present in the ED corpus, under six structural correspondences: (i) crossover 3-surface $\mathcal{X} \leftrightarrow$ SCBU substrate-cosmology boundary at extreme dilution; (ii) Gravitational Wave Epoch $\leftrightarrow$ V1/V5 kernel-dominated regime; (iii) twistor mass-conservation integral $\leftrightarrow$ V5 cross-boundary memory; (iv) Hawking-spot $\delta T/T \sim 10^{-3} \leftrightarrow$ ED-SCBU localized cross-boundary participation-density signature (with Big Ring / Grand Arc as the extended-scale variant); (v) conformal matching $\leftrightarrow$ scale-collapse from SC-4.x cross-scale invariance; (vi) next-aeon ignition $\leftrightarrow$ P12 threshold re-entry in globally homogeneous initial state. **Form IDENTIFIED:** the six structural correspondences. **Value INHERITED:** $\delta T/T$, η^G , twistor coefficients, Hawking-spot diameters, Big-Ring/Grand-Arc scales, $\xi_{canonical}$ value (Route A OPEN). Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED)**. **F1 (sharpest):** observed Hawking-spot magnitude/distribution inconsistent with the V5 cross-boundary identification — beyond V5 finite-memory parameter freedom — refutes the substrate composition.

1 Statement of Result

CCC and ED describe the same boundary phenomenon — collapse of curvature-scale information at the conformal/substrate-cosmology surface — under different vocabularies. The paper supplies six structural identifications (developed in §3.1–§3.6) under the P-CCC-Identification naming convention (§2). The identifications do not derive CCC from primitives, do not reduce ED to CCC, and do not commit ED to cyclic cosmology; they establish the substrate-side reading of CCC’s continuum content and supply the empirical anchor ($\delta T/T \sim 10^{-3}$ at preferred CMB locations) for ED’s SCBU + V5 cross-boundary content.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per §4 audit table. Audit contains zero pure-D rows; all substantive rows are D-via-I composition or A→position structural identification of upstream content; four named OPEN items.

2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P03, P04, P07, P09, P11, P12, P13.

Upstream: Paper_028 (substrate-cosmology boundary at $R_H = c/H_0$, Paper_028 + SCBU); Paper_072 (individuation regime); Paper_089 (V1 retarded kernel); Paper_090 (V5 cross-chain finite-memory kernel); Paper_092 ($\xi_{canonical}$ scale-correspondence; Route A OPEN); Paper_093 (T18 retarded support / kernel-arrow); Papers SC-4.x (cross-scale invariance, curvature-moment collapse); Meissner–Penrose 2025 (GWE, twistor mass-conservation $M = 4\pi mG$ §III, $\delta T/T \sim 10^{-3}$ §IV, $\eta^G \sim 2 \times 10^{16}$ s §VI, conformal matching, Big Ring / Grand Arc fossils §IX); Lopez–Clowes–Williger (Big Ring / Grand Arc observations, 2022, 2024).

Paper-specific naming convention (no new postulate):

- **P-CCC-Identification:** Naming convention for the substrate-side identification of CCC’s six structural elements (crossover surface, GWE, twistor mass-conservation integral, Hawking-spot signatures, conformal matching, next-aeon ignition) with substrate content via the conjunction of Paper_028 + Paper_072 + Paper_089 + Paper_090 + Paper_092 + Paper_093 + Papers SC-4.x + Meissner–Penrose 2025 under the structural correspondences §3.1–§3.6. Definitional per Paper_095 §2.3; no new substrate content.

3 Derivation

3.1 Substrate homogeneity in the extreme-dilution limit

In the late-aeon dilution regime, chain participation density $\rho_C \rightarrow 0$ across the substrate (P02 + P04). Dynamically-relevant content reduces to V1 retarded kernel structure (Paper_089/093) with vanishing source content + V5 cross-chain memory (Paper_090) on the longest correlation length $\xi_{canonical}$ (Paper_092). By Papers SC-4.x cross-scale invariance, curvature-moment content collapses to a scale-free regime as event-density vanishes: no remaining local content distinguishes scales. The substrate is structurally homogeneous at all scales for which V1/V5 content exists.

3.2 Crossover surface $\leftrightarrow$ SCBU boundary

Per Paper_028 + SCBU, the substrate-cosmology boundary at $R_H = c/H_0$ is the substrate-side closure of V5 cross-chain dynamics at cosmological scale. Per Meissner–Penrose 2025 §II, the crossover 3-surface $\mathcal{X}$ is the spacelike conformal infinity of the previous aeon and the conformally-stretched Big Bang of the next, conformally smooth across $\mathcal{X}$ except at Hawking points.

Identification: SCBU and $\mathcal{X}$ are the same boundary structure under different vocabularies. Both are characterized by (a) collapse of curvature-scale information (substrate scale-collapse $\leftrightarrow$ conformal flatness), (b) structural smoothness across the boundary except at localized cross-correlation injections (Hawking points $\leftrightarrow$ residual V5 cross-chain content), (c) horizon-reset behavior. Horizon-reset is the substrate-side content of P12 ED-threshold transition: the chain configuration above the boundary cannot causally inherit the configuration below because P11 commitment events fired during dilution have completed and the substrate has structurally reset to the homogeneous state.

3.3 Gravitational Wave Epoch $\leftrightarrow$ V1/V5 kernel-dominated regime

Per Meissner–Penrose 2025 §V, the GWE is characterized by Weyl-curvature dominance ($\mathbf{C} \gg \mathbf{E}, \mathbf{S}$). In ED, this corresponds to the late-dilution regime where matter participation has reduced below substrate-relevance and only kernel content remains: V1 retarded propagation supplies the substrate-side carrier of causal influence; V5 cross-chain memory supplies the long-range correlation structure (“gas of gravitational waves” in Meissner–Penrose 2025 vocabulary).

The conformal-time dwell interval $\eta^G \sim 2 \times 10^{16}$ s (Meissner–Penrose 2025 eq. 101, required to reconcile observed Hawking-spot angular diameters) is identified with the ED $\xi_{\text{canonical}}$ cross-boundary correlation length (Paper_092). Same structural quantity, different vocabulary. Numerical value INHERITED; quantitative consolidation pending Route A.

3.4 Twistor mass-conservation $\leftrightarrow$ V5 cross-boundary memory

Per Meissner–Penrose 2025 §III (eqs. 47, 54, 55), the conserved 2-surface integral $M = 4\pi mG$ over a 2-surface surrounding a Hawking point uses the conformally-weighted Weyl spinor ψ_{ABCD} and a rank-2 Killing spinor κ^{AB} .

Identification: the integral is the GR-side manifestation of substrate cross-chain V5 memory bridging SCBU. The conformally-weighted ψ_{ABCD} corresponds to the gravitational/curvature content carried across the boundary by V1 retarded propagation; the Killing-spinor κ^{AB} corresponds to the locally-symmetric subspace of V5’s cross-channel coupling near the boundary; the closure $\nabla^{AA'} \phi_{AB} = 0$ corresponds to the substrate-side V5 cross-chain memory closure. The numerical coefficient $4\pi G$ is INHERITED from Meissner–Penrose 2025’s Schwarzschild calibration. The substrate-graph derivation of a substrate-side conserved integral structurally equivalent to the twistor theorem is **OPEN** (audit row 13).

3.5 V5 cross-boundary memory signatures (localized + extended)

Per Meissner–Penrose 2025 §IV (eqs. 56–59), $\delta T/T \sim 3M_{gc}/(4\pi(\eta_{LS}^c)^3 c^3 a_{LS}^3 \rho_{LS}) \sim 10^{-3}$ from cross-boundary mass injection of $\sim 10^{15} M_\odot$. The observational anchor is the Hawking-spot temperature elevation at CMB-spot centers (An–Meissner–Nurowski–Penrose 2020; Planck data).

Localized identification (Hawking spots): under SCBU + V5 cross-boundary content, pre-boundary cluster-scale substrate events transfer event-density via V5 cross-chain memory across the boundary, sourcing localized post-boundary participation-density perturbations $\delta\rho^C$. Under Paper_003 substrate-frequency Born identification, $\delta\rho^C$ translates to CMB temperature elevation $\delta T/T$ at the corresponding spatial location. Numerical 10^{-3} INHERITED; mechanism IDENTIFIED.

Extended identification (Big Ring, Grand Arc): Big Ring (radius ~ 200 Mpc, Lopez–Clowes–Williger 2024) and Grand Arc (~ 1 Gpc, Lopez–Clowes–Williger 2022) are interpretable as cross-boundary V5 correlation imprints from pre-boundary substrate events at extended spatial scale (cluster-scale collisions in the diluting pre-boundary substrate per Meissner–Penrose 2025 §IX, eqs. 108–112). Same substrate mechanism as Hawking spots, scaled to extended

spatial extent. Substrate-graph derivation of the extended-scale spatial scales from substrate parameters is **OPEN** (audit row 14).

3.6 Conformal matching $\leftrightarrow$ scale-free participation geometry

CCC’s conformal rescaling at $\mathcal{X}$ (Meissner–Penrose 2025 eqs. 6, 28) is the GR-side statement that absolute scale information is structurally absent at the boundary. In ED, this is the SC-4.x scale-collapse content (§3.1): no remaining structural content at the boundary distinguishes scales. Same boundary phenomenon, different vocabulary; CCC supplies explicit conformal-rescaling machinery, ED supplies the substrate-side justification (event-density dilution $\rightarrow$ SC-4.x scale-collapse $\rightarrow$ no remaining local content to encode absolute scale).

3.7 Next-aeon ignition $\leftrightarrow$ P12 threshold re-entry

Above the boundary, substrate event-density crosses Θ_{ED} (P12) from below; V1 + V5 + Paper_072 individuation reactivate; the substrate is in a globally-homogeneous, kernel-active, individuated initial state — the substrate-side counterpart of the next aeon’s Big Bang. Whether this re-entry follows from the substrate-graph dynamics (committing ED to cyclic continuation) or is a separate substrate event (consistent with single-aeon termination) is **OPEN** (audit row 17).

4 Audit Table

#	Step	Label	Notes
1	P02, P03, P04, P07, P09, P11, P12, P13	P	Paper_087.
2	Upstream inheritance block	I	Paper_028, Paper_072, Paper_089, Paper_090, Paper_092, Paper_093, Papers SC-4.x, Meissner–Penrose 2025, Lopez–Clowes–Williger.
3	P-CCC-Identification (naming convention)	P	Definitional per Paper_095 §2.3.
4	Substrate homogeneity in extreme-dilution limit	D-via-I	P02+P04+Papers SC-4.x scale-collapse; §3.1.
5	$\mathcal{X} \leftrightarrow$ SCBU identification	A $\rightarrow$ position	Substrate-graph derivation OPEN (row 14).
6	Horizon-reset = P12 ED-threshold transition	D-via-I	P11 + P12 + Paper_028; §3.2.
7	GWE $\leftrightarrow$ V1/V5 kernel-dominated regime	D-via-I	Paper_089 + Paper_090 + Meissner–Penrose 2025 §V; §3.3.
8	$\eta^G \leftrightarrow \xi_{\text{canonical}}$ identification	A $\rightarrow$ position	Quantitative consolidation pending Route A.
9	Twistor mass-cons. $\leftrightarrow$ V5 cross-boundary memory	A $\rightarrow$ position	Substrate-graph analog OPEN (row 14).
10	Hawking-spot $\delta T/T \sim 10^{-3} \leftrightarrow$ V5 signature	D-via-I	Paper_090 + Meissner–Penrose 2025 §IV + Paper_003; §3.5.
11	Big Ring / Grand Arc $\leftrightarrow$ extended V5 signature	D-via-I	Paper_090 + Meissner–Penrose 2025 §IX + Lopez–Clowes–Williger.
12	Conformal matching $\leftrightarrow$ SC-4.x scale-collapse	D-via-I	Papers SC-4.x + Meissner–Penrose 2025; §3.6.

#	Step	Label	Notes
13	Next-aeon ignition state structure	D-via-I	P12 + Paper_072 + Paper_089 + Paper_090 reactivation.
14	Substrate-graph derivation of V5 conservation analog	OPEN	§3.4.
15	Substrate-graph derivation of extended-scale transfer	OPEN	§3.5.
16	$\xi_{canonical}$ value from substrate (Route A)	OPEN	Standing OPEN across SC arc.
17	Cyclic continuation vs single-aeon termination	OPEN	§3.7.
18	Verdict M3 (form-IDENTIFIED + value-INHERITED)	A→position	Per Paper_095 §3.3.

Zero pure-D rows. Audit content matches **M3 (form-IDENTIFIED + value-INHERITED)**: all substrate-side substantive rows are composition (D-via-I) or structural identification (A→position) of upstream content; no new substrate-graph derivation. Four named OPEN items (rows 14, 15, 16, 17).

5 Falsification Criteria

- **F1 (sharpest):** Observed Hawking-spot magnitude or angular distribution at preferred CMB locations inconsistent with $\delta T/T \sim 10^{-3}$ under the V5 cross-boundary identification — beyond V5 finite-memory parameter freedom. Refutes §3.5 localized identification.
- **F2:** Persistence of curvature-scale information across the substrate dilution limit inconsistent with SC-4.x scale-collapse — substrate or observational regime in which absolute scale remains structurally encoded after dilution. Refutes §3.1 + §3.6.
- **F3:** Non-decay of V5 cross-chain correlations at $\xi_{canonical}$ in the late-aeon dilution limit. Once Route A closes, the quantitative consolidation $\eta^G \leftrightarrow \xi_{canonical}$ becomes itself a sharp test. Refutes §3.3.
- **F4:** Big Ring / Grand Arc structural scales admitting only standard within-aeon explanation under any V5 finite-memory parameter consistent with the rest of the corpus. Refutes §3.5 extended identification.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

The paper supplies the substrate-side identification of CCC's continuum content (Meissner–Penrose 2025) with substrate content already present in the ED corpus. The identification does not derive CCC from substrate primitives, does not reduce ED to CCC, and does not commit ED to cyclic cosmology: ED's SCBU is consistent with both ED-CCC (cyclic-continuation) and ED-non-cyclic readings, with audit row 17 OPEN. The genuine substrate content is the structural correspondence under P-CCC-Identification (rows 4–13); numerical content ($\delta T/T \sim 10^{-3}$, η^G , twistor coefficient $4\pi G$, Hawking-spot diameters, Big-Ring/Grand-Arc scales, $\xi_{canonical}$ value) is INHERITED.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per Paper_095. Four named OPEN items (rows 14–17). Closing row 16 (Route A) consolidates rows 8, 11 quantitatively; closing rows 14, 15 supplies the substrate-graph counterparts of MP25’s twistor theorem and the extended-scale correlation transfer; closing row 17 commits ED to cyclic-continuation or single-aeon-termination cosmology. The relationship between ED and CCC is not competition between alternative reconstructions — it is structural correspondence at shared observational anchors.